

Lecture - 1

Linear Equations

An equation of the form

(A) $a_1x_1 + \dots + a_nx_n = b$, where $a_1, \dots, a_n$

and b are numbers and $x_1, \dots, x_n$ are variables is called a linear equation.

If $\alpha_1, \alpha_2, \dots, \alpha_n$ are numbers such that

$$a_1\alpha_1 + \dots + a_n\alpha_n = b$$

then $x_1 = \alpha_1, \dots, x_n = \alpha_n$ is a solution of the Equation (A).

Example 1

$5x = 2$ is a linear equation having $x = \frac{2}{5}$ as a solution. Note that $x = \frac{2}{5}$ is the only solution!

Example 2

$3x - 2y = 6$ is a linear equation. Clearly $x = 2$ and $y = 0$ is a solution of this equation. Also, $x = 0$ and $y = -3$ is also a solution of this equation.

What are all the solutions of $3x - 2y = 6$?

* If $x = \alpha$, where α is any number then $x = \alpha, y = \frac{6 - 3\alpha}{2}$ is a solution of the equation $3x - 2y = 6$.

* If $x = \alpha$ and $y = \beta$ is a solution of $3x - 2y = 6$ then

$$\beta = \frac{6 - 3\alpha}{2}.$$

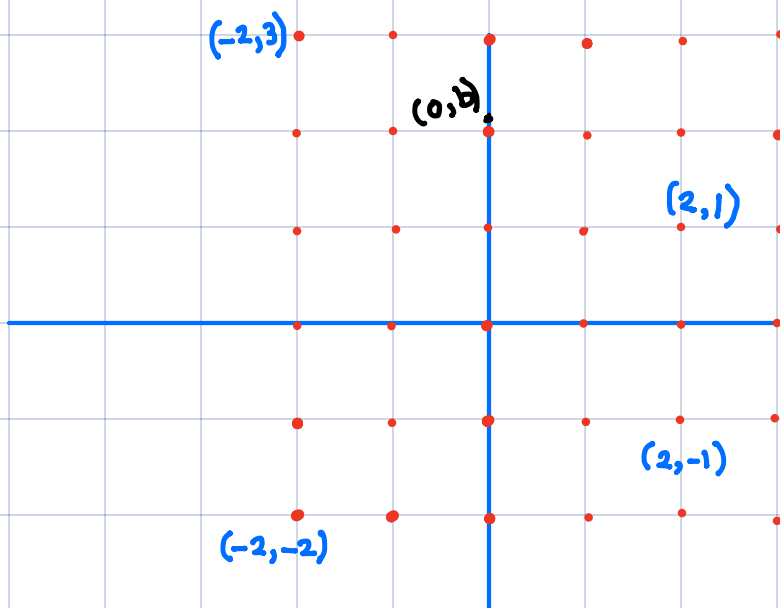
Therefore, for any number α , $x = \alpha$ and $y = \frac{6 - 3\alpha}{2}$

Provide all solutions of the Equation $3x - 2y = 6$.

$\mathbb{R}$ - Real numbers, $\mathbb{Q}$ - Rational numbers, $\mathbb{Z}$ - integers.

Euclidean Plane

$$\mathbb{R}^2 = \mathbb{R} \times \mathbb{R} = \{(a, b) \mid a \in \mathbb{R}, b \in \mathbb{R}\}$$



X-Coordinate

(a, b)

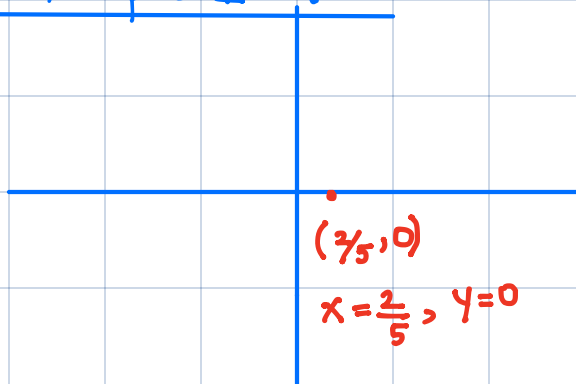
Y-Coordinate

$(a, 0)$

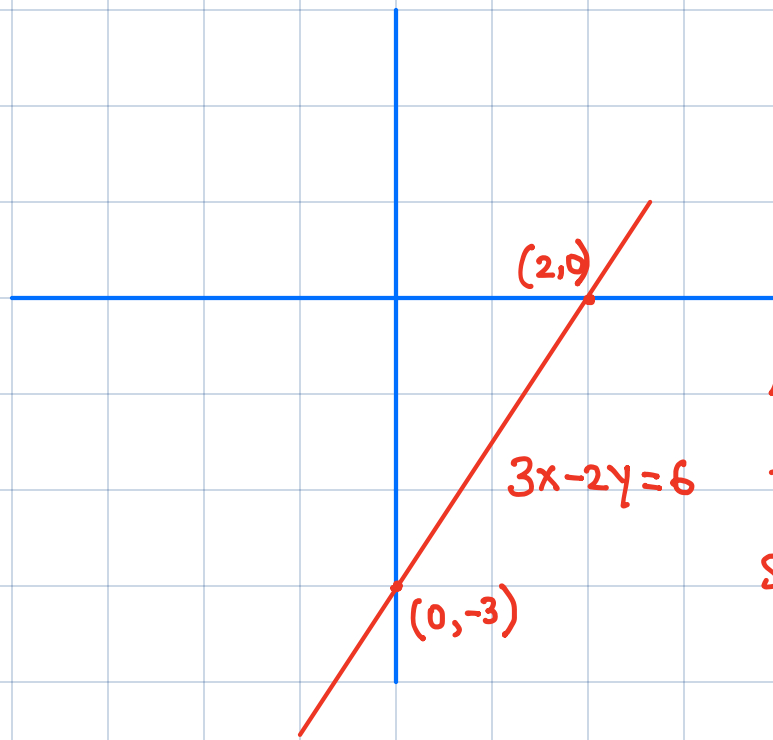
Red dots are
"integer points"

Pictorial Representations of Examples 1 and 2

Example 1



Example 2



All points on
this line are
solutions of
 $3x - 2y = 6$

System of linear Equations

Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

$\vdots$

$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = b_m$$

where $a_{11}, a_{12}, \dots, a_{m1}, a_{m2}, \dots, a_{mn}$ are (real) numbers and $b_1, b_2, \dots, b_m$ are numbers and $x_1, \dots, x_n$ are variables form a "system of equations".

If $d_1, \dots, d_n$ are numbers such that

$$a_{11}d_1 + a_{12}d_2 + \dots + a_{1n}d_n = b_1$$

$\vdots$

$$a_{m1}d_1 + a_{m2}d_2 + \dots + a_{mn}d_n = b_m$$

then $x_1 = d_1, x_2 = d_2, \dots, x_n = d_n$ is a solution of (B)

Example 3

What are the solutions of the system

$$\left. \begin{array}{l} 3x - 2y = 6 \\ x - y = 0 \end{array} \right\} \text{..... } \textcircled{C}$$

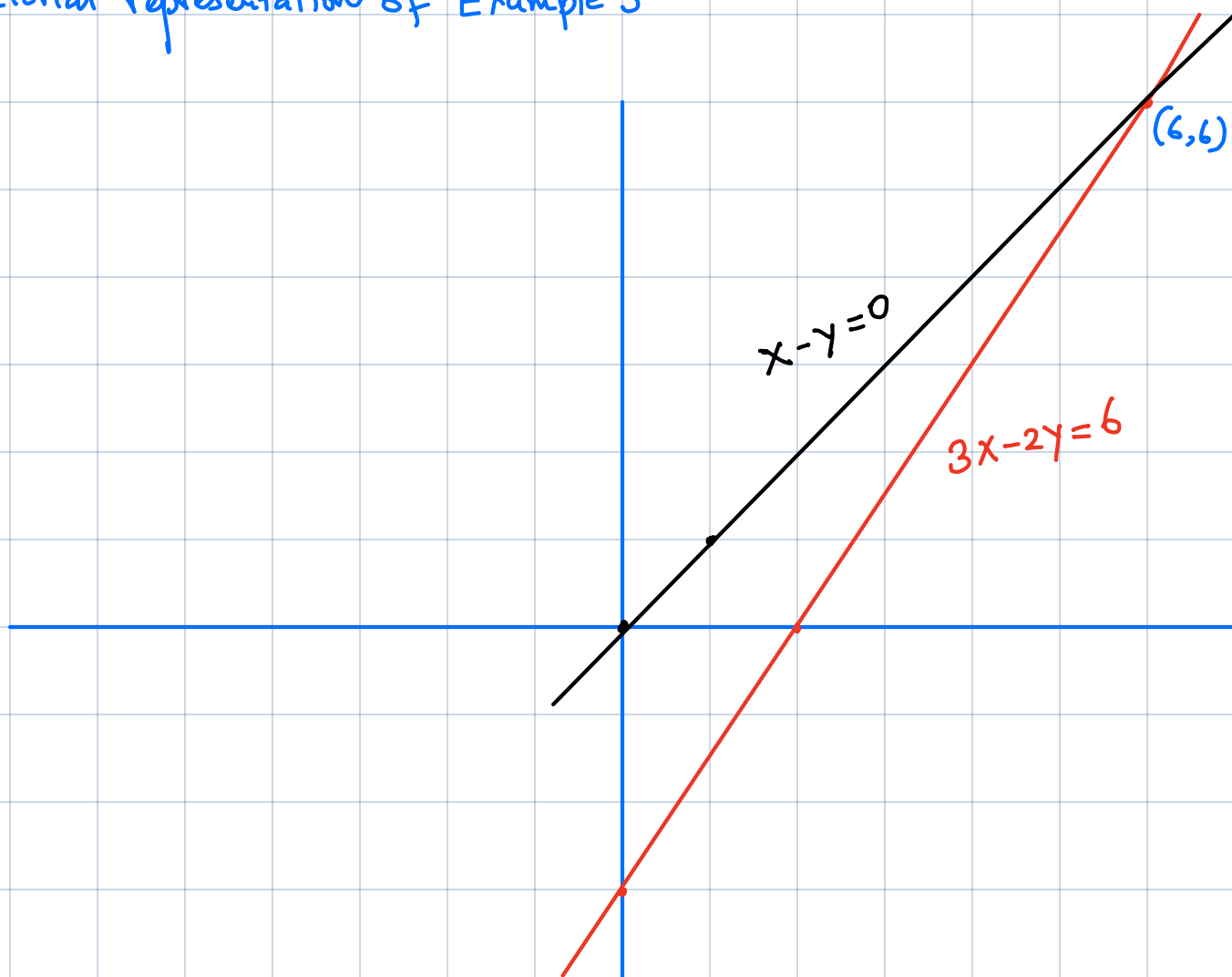
If $x = \alpha$, $y = \beta$ is a solution then $x - y = 0$ implies $x = y$, that is, $\alpha = \beta$.

Now $x = \alpha$, $y = \beta$ is a solution of $3x - 2y = 6$ implies

$$3\alpha - 2\beta = 6, \text{ that is, } 3\alpha - 2\alpha = 6.$$

Then $\alpha = 6$ and $\beta = 6$. Thus $\boxed{x = 6, y = 6}$ is a solution of $\textcircled{C}$

Pictorial representation of Example 3



What are other possible solutions of (C)?

* $x=b \neq y=b$ is the only solution!

Equation of a line

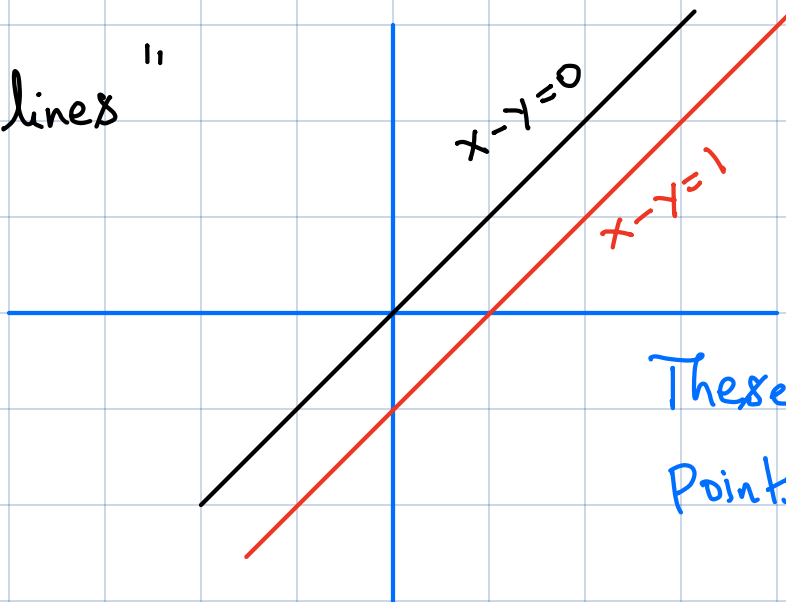
The set of all solutions of the equation

$$ax - by = c, \text{ where } a, b, c \text{ are real numbers}$$

with either $a \neq 0$ or $b \neq 0$ is called a line in $\mathbb{R}^2$.

* Do any two lines intersect?

"Parallel lines"



points on black line
 $\{(a, a) / a \in \mathbb{R}\}$

Points on red line
 $\{(a, a-1) / a \in \mathbb{R}\}$

These two sets have NO
Points in Common!

Projective Geometry

"Parallel line meets at
infinity"

